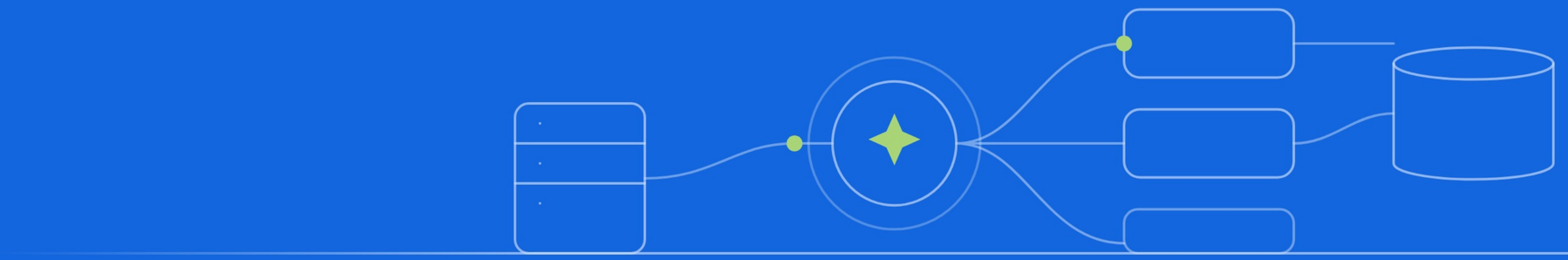
Beside
ERP Solutions

UNIQUE POLYMER · In-house AI system · machine, models and MCP

In-house AI

Company data stays in the company · ask the system anything



Agenda topics

1 Brief and architecture

Needs & Architecture

- The 5 jobs the customer needs
- System overview and MCP Gateway

2 MCP and SAP B1

MCP & SAP B1

- What is MCP · 7 MCP servers
- Connect SAP B1 read-only

3 Models and screens

Models & Screens

- Model set on one machine
- Sample: ask for a report from SAP B1

4 Machine, security and plan

Hardware & Plan

- PGX Spark vs Mac Studio
- Permissions · audit log · pilot plan

The customer needs 5 things from AI, and data must not leave the company

1 Chat Q&A
Chat

- Ask about work every day
- Search the company's documents and manuals

2 Reports
Report

- Ask in plain language
- Get figures from SAP B1 and Beside, with tables

3 Image generation
Image

- Illustrations for documents and presentations
- Images never leave the company

4 File translation
Translate

- PDF / Word / Excel
- Thai–English, files never leave the machine

5 Slide creation
Slides

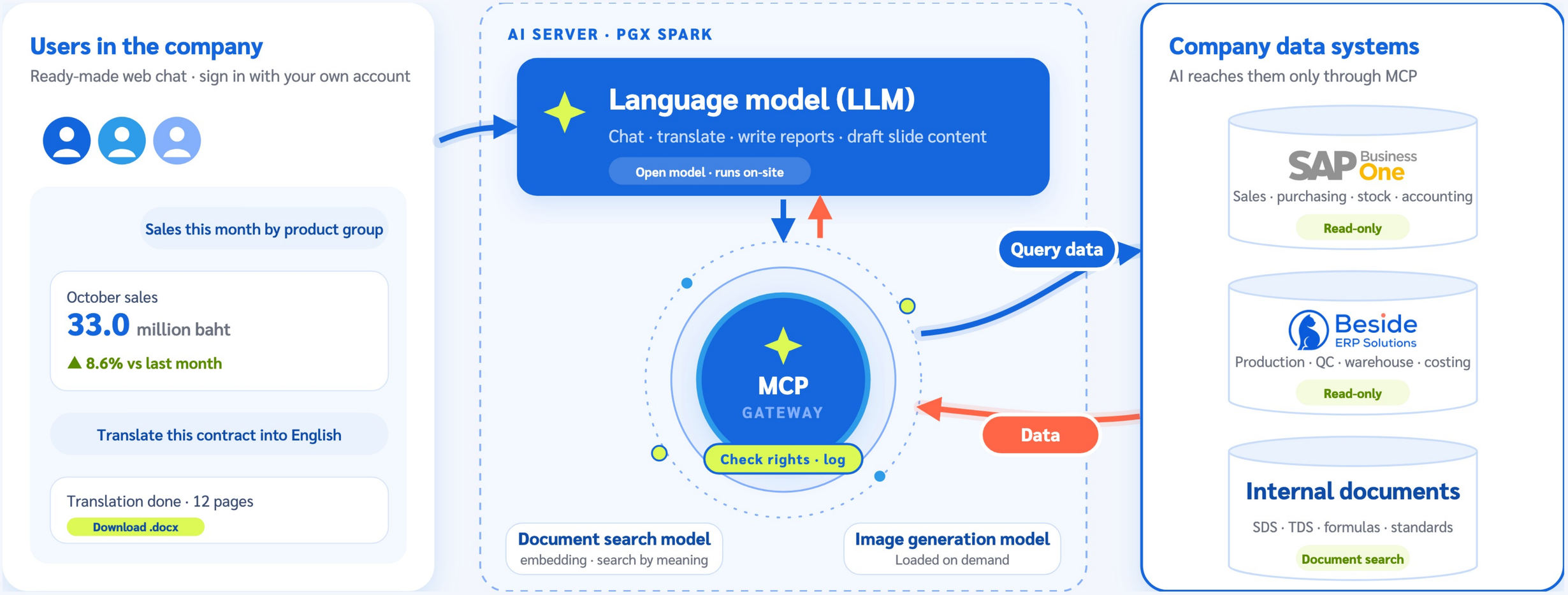
- AI drafts the content
- Uses the company's PowerPoint template

Key condition **All data stays on company machines — nothing is sent to external AI services**

Why on-premise: sales, cost, production formulas and customers are trade secrets. The customer wants to control where data lives and who can access it.

Architecture · system overview

The LLM never touches the database directly — always through the MCP Gateway

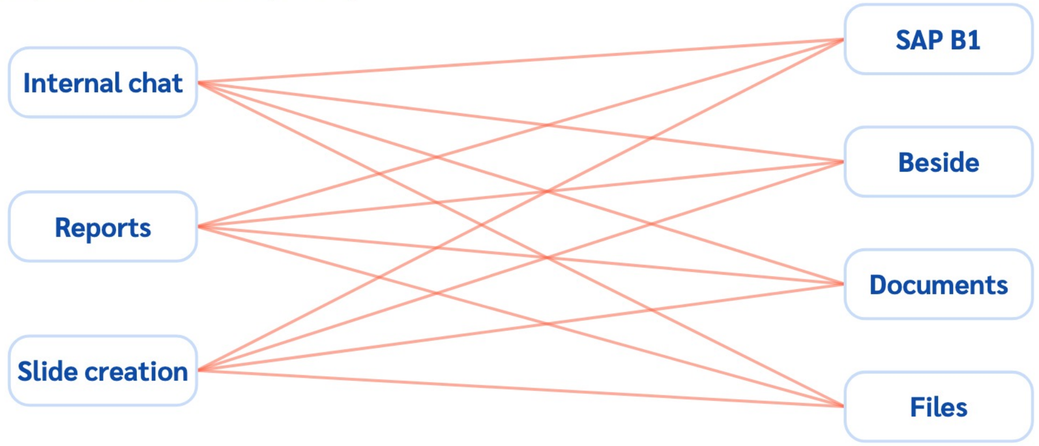


Key point: the model can only use the tools that are enabled, and sees only the data the asker is allowed to see — even if the model misreads an instruction or is tricked. · **Chat screen:** use a ready-made open-source program (e.g. Open WebUI, LibreChat) — no new app to develop; final choice made during the pilot

What is MCP — a standard plug between AI and the company's systems

Without MCP

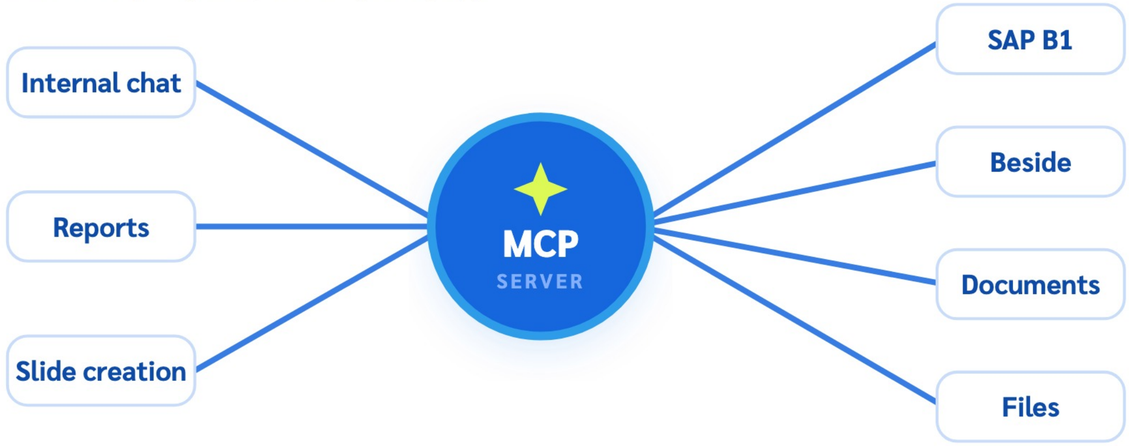
Each pair must be wired separately



3 × 4 = 12 connectors

With MCP

Write once per system, usable by every app



3 + 4 = 7 connectors

1 The LLM requests a tool

Step 1

- e.g. "Show October sales by product group"

2 The MCP server checks

Step 2

- Is the asker allowed? Is this tool permitted?
- Limit the scope of data

3 Only permitted data goes back

Step 3

- The LLM turns it into an answer
- States the sources it pulled from

7 MCP servers — read-only first

MCP server	What it does	Mode	Connects to
SAP B1	Sales, purchase orders, stock, customers, items	Read-only	B1 SDK + defined SQL views
Beside	Production orders, cost per order, machines, scrap	Read-only	Beside database
Internal documents	Search SDS · TDS · formulas · quality standards	Read-only	Local files + vector database
Files & translation	Read PDF / Word / Excel, then translate or summarise	Read-only	Files the user uploads
Image generation	Create images from a description, returned as files	Creates new files	On-site image model · queued
Slides & reports	Fill the company's .pptx / .xlsx templates	Creates new files	Company template
SAP B1 · write	Create documents, e.g. purchase requisition (PR)	Later phase · human confirms	Via SDK only

Principle: start with read-only tools only — changing or creating documents in the system is a later phase, and always needs a person to confirm.

Connecting SAP B1 — read-only, with permissions per user

SAP Business One

Core system for accounting, sales and purchasing

Sales
Sales Order
Delivery · invoice

Purchasing
PR · PO
Goods receipt · payables

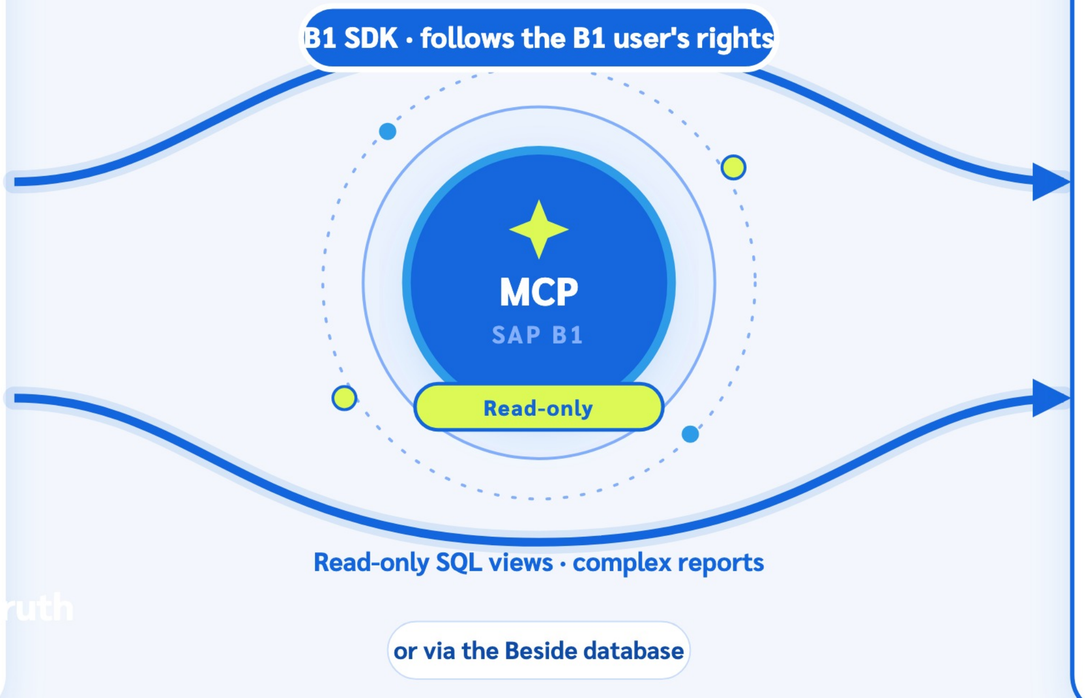
Stock
Inventory
Lot · warehouse

Master data
Items · customers
Vendors

BOOK OF RECORD

SAP B1 = the accounting source of truth

AI can read, never writes directly



AI Server

Language model + MCP Gateway

Language model (LLM)
Turns the question into a tool call, then summari

Data dictionary (semantic layer)
"Sales" = which field · are cancelled invoices excluded?
Explain the company's UDFs to the AI

Information from the company
① System: SAP B1 (database to be confirmed)
② Connect through the B1 SDK
③ AI chat users ≈ 5 (estimate)

SAP licensing — check before connecting

One shared account for many askers may count as indirect access. Have the SAP consultant confirm the contract.

No direct writes to the database

If writing is added later, it must go through the SDK with a person confirming.

Test with real questions

Build the data dictionary with the users and compare figures with SAP every time during the pilot.

Model set · 5 jobs on one machine

No single model does everything — use a set of models on one machine

1 Chat / Q&A
Main LLM

- Large open model, e.g. gpt-oss-120B
- Always on

2 SAP / Beside reports
LLM + MCP

- The same model calls tools
- Shows sources and figures to check

3 Internal document search
embedding

- Small semantic-search model
- Uses little memory

4 File translation
Main LLM

- Test quality per language
- Especially Thai

5 Image generation
Separate image model

- Open source, FLUX / SDXL family
- Loaded when used, so it does not compete for memory

6 Slide creation
LLM + template

- AI writes the content, a program builds the file
- Uses the company's template

128 GB
Total memory of the machine, shared by all models

≈ 60 GB
Size of the main language model (gpt-oss-120B, 4-bit compressed)

≈ 59 words/sec
Response speed measured by an outside tester on the same machine

Note: the speed figure comes from third-party testing and must be re-measured on real work during the pilot. A model running on a single machine is lower in quality than top-tier AI services, especially on complex tasks.

Sample screen · ask for a report from SAP B1

Ask in plain language, get a table with sources

Sample data

In-house AI · Unique Polymer Sample screen

Compare October with September sales by product group, as a table

`sap.sales_by_group` read-only `Rights: Sales ✓`

Total October sales**33.0 million baht**Up 8.6% from September (30.4 million baht), mostly from the bioplastic group

Product group	Sep (million baht)	Oct (million baht)	Change
White masterbatch	11.1	12.4	+11.7%
Slip masterbatch	7.3	6.8	-6.8%
UV-protection masterbatch	4.6	5.2	+13.0%
Anti-static masterbatch	3.5	3.9	+11.4%
BIONEX™ bioplastic	3.9	4.7	+20.5%
Total	**30.4**	**33.0**	**+8.6%**

Source: closed sales invoices (cancellations excluded) · Would you like a summary slide?

Create reports

Export to Excel or PDF in the company template

Translate files

Drag a PDF / Word file into the chat and get the translated file back

Generate images

Describe the image you need and get an image file for documents

Create slides

AI drafts the content as PowerPoint in the template

Hardware · comparing 2 options

PGX Spark is the best fit for this job

	PGX Spark (NVIDIA GB10)	Mac Studio M5 Max
What it looks like	 Sample GB10 machines from HP, Lenovo, MSI (also Dell, Acer and others)	 Mac Studio
Memory	128 GB	128 GB
Approx. price	฿200,000–263,000 4TB model, varies by brand and shop	฿191,400
Response speed (memory)	273 GB/s · enough for chat	Higher than Spark
LLM + image software	Complete, standard NVIDIA CUDA	Possible, but fewer options and slower image generation
Multiple concurrent users	Better (server-style queuing supported)	Supports fewer than the PGX Spark
Form factor	Linux server in the network room	Desktop machine
Conclusion	**Choose this one** — suits many jobs and many users	Secondary option

Note: Spark prices checked at Thai shops on 1 Nov 2026 and differ by tens of thousands of baht between brands and shops. Mac Studio price is from the Apple Thailand site. Memory capacity (128 GB) is the main limit on model choice. · Machine photos: NVIDIA and Apple.

Privacy and permissions — controlled in one layer

Rights follow the B1 user
Per-user access

- The asker sees only the data they are allowed to see
- No single shared account (check licensing terms)

Read-only first
Read-only first

- Every SAP and Beside tool is read-only
- Changing data is a later phase, with a person confirming

Data stays in the company
On-premise

- Models and data live on internal machines
- Nothing goes to external AI services

Every use is logged
Audit log

- Who asked what, what data was pulled, and when
- Can be audited afterwards

Prompt-injection defence
Guardrails

- Documents or emails may hide instructions that trick the AI
- Limit the enabled tools, and require a person to confirm risky ones

Quality check before launch
Quality gate

- Test with the company's real questions
- Compare report figures with SAP every time during the pilot

Principle **Security lives in the MCP Gateway — it does not depend on the model "understanding correctly"**

Start with 1 machine, measure for real, then expand

1 machine · small user group

Pilot phase

- Install 1 PGX Spark
- Connect SAP B1 and Beside read-only
- Let a small group ask real questions

Scale to the number of users

Production phase

- Open to the whole department
- Tune tools and the data dictionary from test results
- Decide whether to add machines

Add capacity

Expansion phase

- Link 2 Sparks, or
- a high-end NVIDIA GPU workstation
- Add write-data tools (with human confirmation)

What we measure during the pilot

Accuracy

Do report figures match SAP B1?

Response time

Is it fast enough for real use?

Thai language

How well does it translate and write reports?

Real usage

How many people, how often, for what?

Machine sizing: about 5 AI chat users — 1 machine is enough to start (estimate, to be confirmed in the pilot) · timeline and price are in the quotation

4 items we need to size the machine and start the pilot

1 Number of users

Question 1

- About 5 AI chat users — please confirm · how many at peak?
- 5 users: 1 machine is enough to start · licensing check

2 The first question to ask

Question 2

- Sales, stock, cost or production?
- Used to pick the first tool to build

3 Image generation

Question 3

- What are images for: documents, advertising, product photos?
- Used to choose the image model

4 Server room

Question 4

- Is there a server room · power, cooling, internal network?
- Used to prepare where the machine goes

Next step Confirm information → set up the machine and connect systems → pilot with a small group



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